

Prototype Screens (Lo-fi Wireframes)

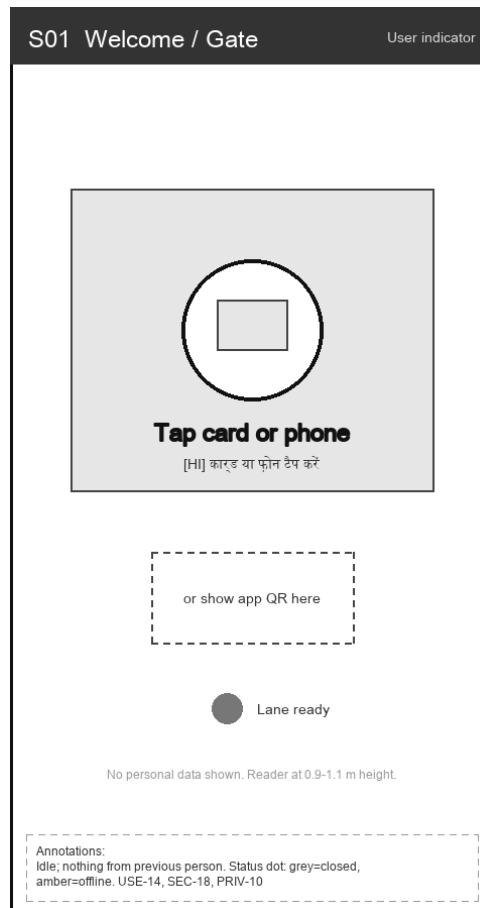
20 screens — per-screen specifications in Prototype_Specification.pdf

CSE 347/652 | DES 306/525 Usable Security and Privacy
IIIT-Delhi · Monsoon 2026 · Instructor: Dr. Arun Balaji Buduru

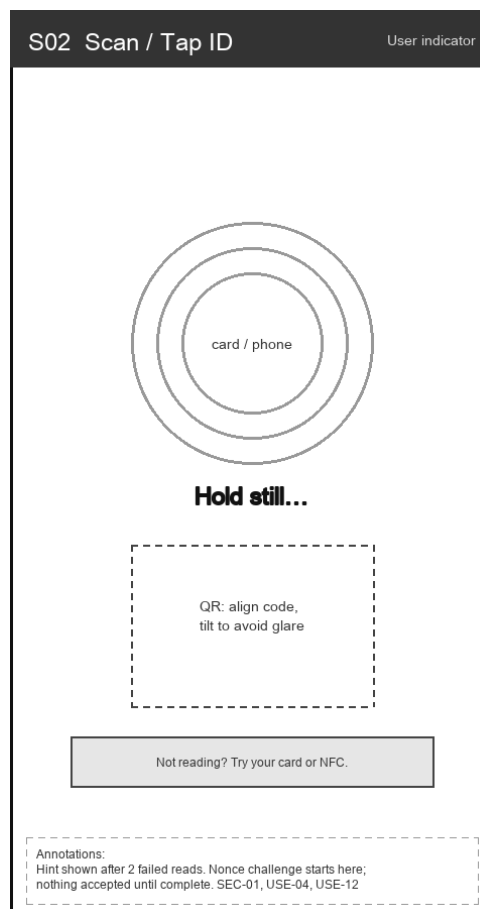
Phase 1: User Studies to evolve Requirements, Design and Workflows
Submission: 4 October 2026

Team: [INSERT TEAM MEMBER NAMES AND ROLL NUMBERS]

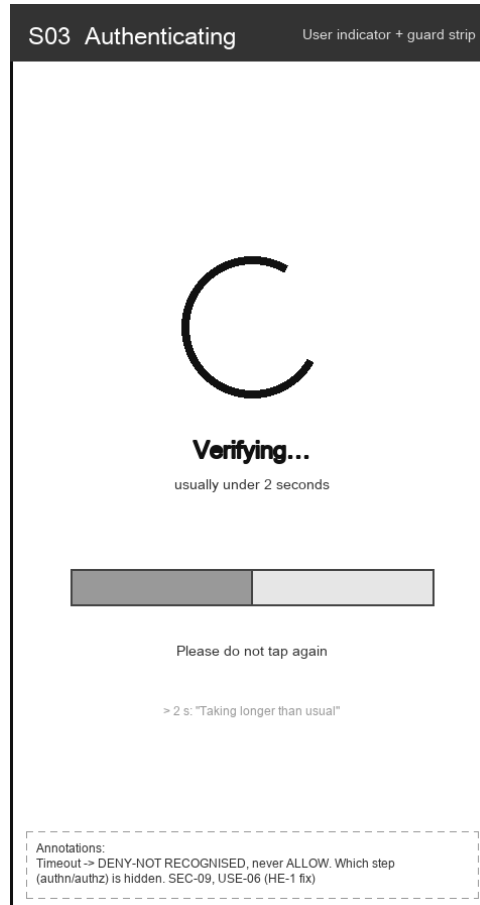
Primary research status: instruments complete; fieldwork data not yet collected. All placeholders are marked [DATA REQUIRED]. No primary data has been fabricated.



S01 — Welcome Gate



S02 — Scan Tap



S03 — Authenticating



S04 — Auth Success

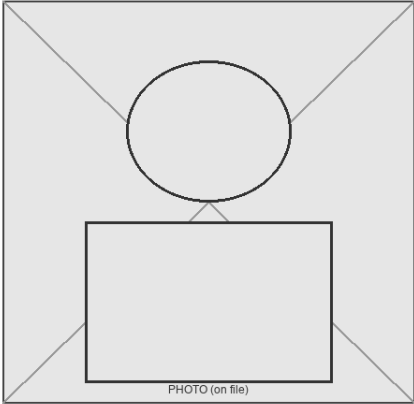
S05 Authorization Success

Guard tablet 10"

• ONLINE | list 00:04 ago | Reader OK | Alerts 0

✓ ALLOWED

GREEN



Aarav

STUDENT

Authenticated and authorized for this gate.
Confirm the face matches the photo.
Method:  card

NEXT STEP

Let the person pass.
If the face does NOT match → press [Mismatch].

Mismatch ✕

Gate: Main • 11:42:07 • Hidden: full name, roll no., programme, hostel, phone, credential ID

Annotations:
 No [Allow] exists. Photo from server, not card. Mismatch needs confirm-tap (HE-12: first tap turns user indicator red). FR-12, SEC-08, USE-05, PRIV-03

S05 — Authorization Success

S06 Access Denied — Not Recognised

Guard tablet 10"

• ONLINE | list 00:04 ago | Reader OK | Alerts 0

✕ NOT RECOGNISED

RED

No identity known
(authentication failed)

The reader could not verify this card or code.
This can happen with damaged, copied or non-campus credentials.

NEXT STEP

Do not admit.
Direct the person to the Security Office.
If they insist → [Call supervisor].

Call supervisor

Done

Gate: Main • 11:42:07 • Hidden: full name, roll no., programme, hostel, phone, credential ID

Annotations:
 Authentication failure: no identity. Guard never judges card authenticity. User side: generic red message. S06b variant (NOT PERMITTED here/how) is AMBER with photo. SEC-01, SEC-04, USE-03, HE-15 wording

S06 — Access Denied Not Recognised

4

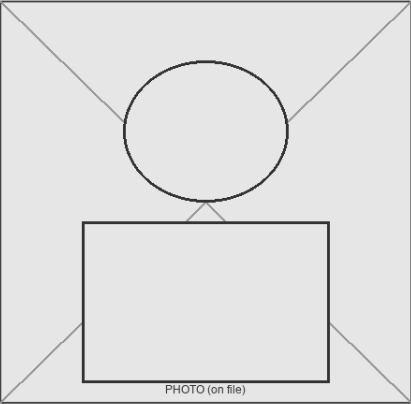
S07 Credential Expired

Guard tablet 10"

• ONLINE | list 00:04 ago | Reader OK | Alerts 0

! EXPIRED

AMBER



Meher

STUDENT

Credential validity has ended.
Identity is known; entry not authorized.

NEXT STEP

Do not admit.
Direct to the ID cell for renewal.
Person says they are a current member → [Call supervisor].

Call supervisor

Done

Gate: Main • 11:42:07 • Hidden: full name, roll no., programme, hostel, phone, credential ID

Annotations:

Authorization failure, identity known. Student gets private explanation in app. User side: 'Credential expired — renew at ID cell'. SEC-03, USE-03

S07 — Credential Expired

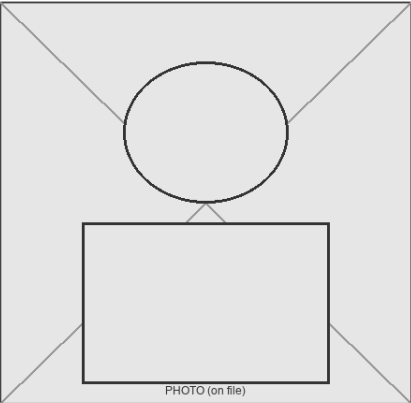
S08 Credential Revoked

Guard tablet 10"

• ONLINE | list 00:04 ago | Reader OK | Alerts 0

× REVOKED

RED



Meher

STUDENT

This credential has been revoked.
(Reason is not shown to gate staff.)

NEXT STEP

Do not admit.
Direct to the Security Office.
Do not discuss the reason.

Call supervisor

Done

Gate: Main • 11:42:07 • Hidden: full name, roll no., programme, hostel, phone, credential ID

Annotations:

Reason withheld (lost vs disciplinary). Flagged automatically. User side identical to S06 for bystanders. SEC-02, SEC-08, DP-13

► Event flagged for supervisor review

S08 — Credential Revoked

S09 Suspicious Credential

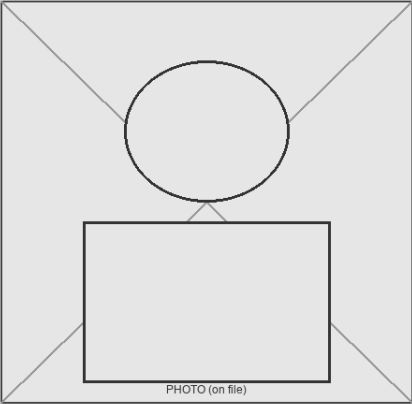
Guard tablet 10"

• ONLINE | list 00:04 ago | Reader OK | Alerts 0

×

SUSPICIOUS — SUPERVISOR ALERTED

RED



Aarav

STUDENT

Repeated failed attempts or face mismatch recorded.
Supervisor has already been notified.

NEXT STEP

Do not admit.
Keep the person at the gate if safe to do so.
Wait for supervisor. Supervisor will clear this alert.

Call supervisor

Done

Gate: Main • 11:42:07 • Hidden: full name, roll no., programme, hostel, phone, credential ID

Annotations:

SEC-13 threshold or [Mismatch]. Reader rate-limited. Photo shown only if identity known. TH-02/04/06

► Event flagged for supervisor review

S09 — Suspicious Credential

S10 Reader Failure

Guard tablet

• ONLINE | list 00:04 ago | Reader OK | Alerts 0 | Reader: FAULT

i

READER OFFLINE — LANE 1

GREY

Cause: self-test failed / tamper detected / no power
This lane is disabled by the server until checked. The reader holds no secret keys.

NEXT STEP

Direct people to Lane 2 (reader OK).
If no other lane is available → start Manual Verification Mode (supervisor PIN).

Report reader problem

Start Manual Verification Mode (PIN)

Call supervisor

Technical diagnostics are available to the auditor, not here.

Annotations:

Reader disabled server-side; tamper logged; cert revocable. Manual mode requires supervisor. SEC-18, FR-14, FR-19, USE-06

S10 — Reader Failure

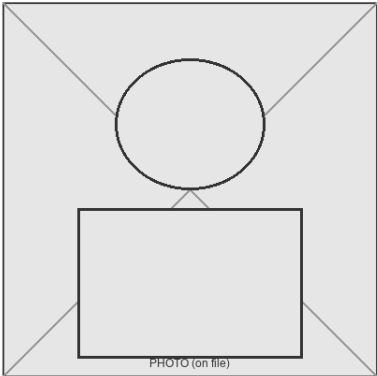
S11 Network Failure (Offline-verified)

Guard tablet

OFFLINE — verifying locally | Last update from server: 1 h 20 m ago (limit 4 h) | Reader OK | Alerts 0

✓ ALLOWED

GREEN



Aarav

STUDENT

Normal checks continue using cached keys and the signed revocation list.
Events are stored on the reader and sent when the connection returns.

IF LIMIT (4 h) IS REACHED
A dialog will require Manual Verification Mode (supervisor PIN).
Until then, work normally.

Mismatch ×

Annotations:
Bounded staleness visible as plain language (HE-2 fix). Automatic transition at limit; no auto-ALLOW. SEC-19, FR-14, USE-06

S11 — Network Failure

S12 Manual Verification / Fallback

Guard tablet (or paper form)

MANUAL VERIFICATION MODE | supervisor: SUP-07 approved start | Reader OK | Alerts 0

All entries in this mode are recorded with your ID and reviewed.

1

Ask for card or app screen

2

Compare face with the card photo

3

Credential shown:

Card

Phone

None

4

Photo check:

Matches

Does not match

5

Supervisor approval:

Request supervisor approval (PIN)

Approve next 10 min (SUP)

6

Result:

Admit / Do not admit (set by supervisor)

! No roster lookup. No personal data typed. Egress is never blocked. [Emergency ingress — supervisor only, reason required]

Annotations:
Weakest mode by design: every decision carries guard + supervisor IDs, flagged. HE-6: collapse to 3 big steps in next iteration. FR-14, FR-19, FR-20, TH-17

S12 — Manual Verification

S13 Visitor Registration (host request)

Host web/app • guard-tablet variant inset

Request a visitor pass

Visitor name

Purpose (category)

Date

Time window

Gate(s)

☒ Main ☐ East ☐ Vehicle

Send pass to

Submit request

Cancel

GUARD VARIANT — Visitor without approval

Host (search by name / department)

Visitor first name

Send approval request to host's phone

Waiting for host... (no entry until approved)

Guard never enters or sees visitor contact details.

Annotations:

Purpose is a category, not free text. Max window enforced (HE-5). Guard variant: no typing beyond host name. FR-10, PRIV-09, SEC-16

S13 — Visitor Registration

S14 Visitor Approval

Approver / host phone

Approval request

Visitor: Dr. Fernandes

Purpose: Academic

Date: 01 Oct 2026

Window: 14:00 — 17:00

Gate: Main

Requested by: Prof. R... (CSE)

Source: gate walk-in • waiting 0:42

Approve

Decline

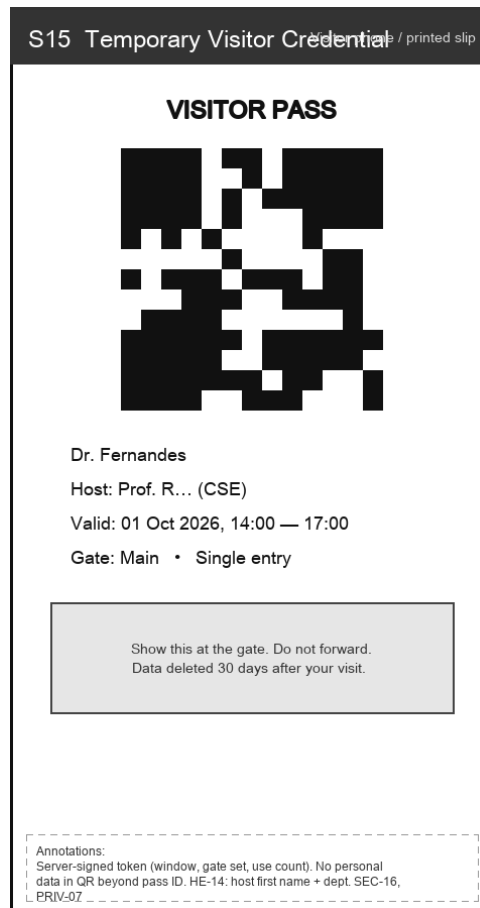
Edit window

Approving creates a signed, single-entry pass.

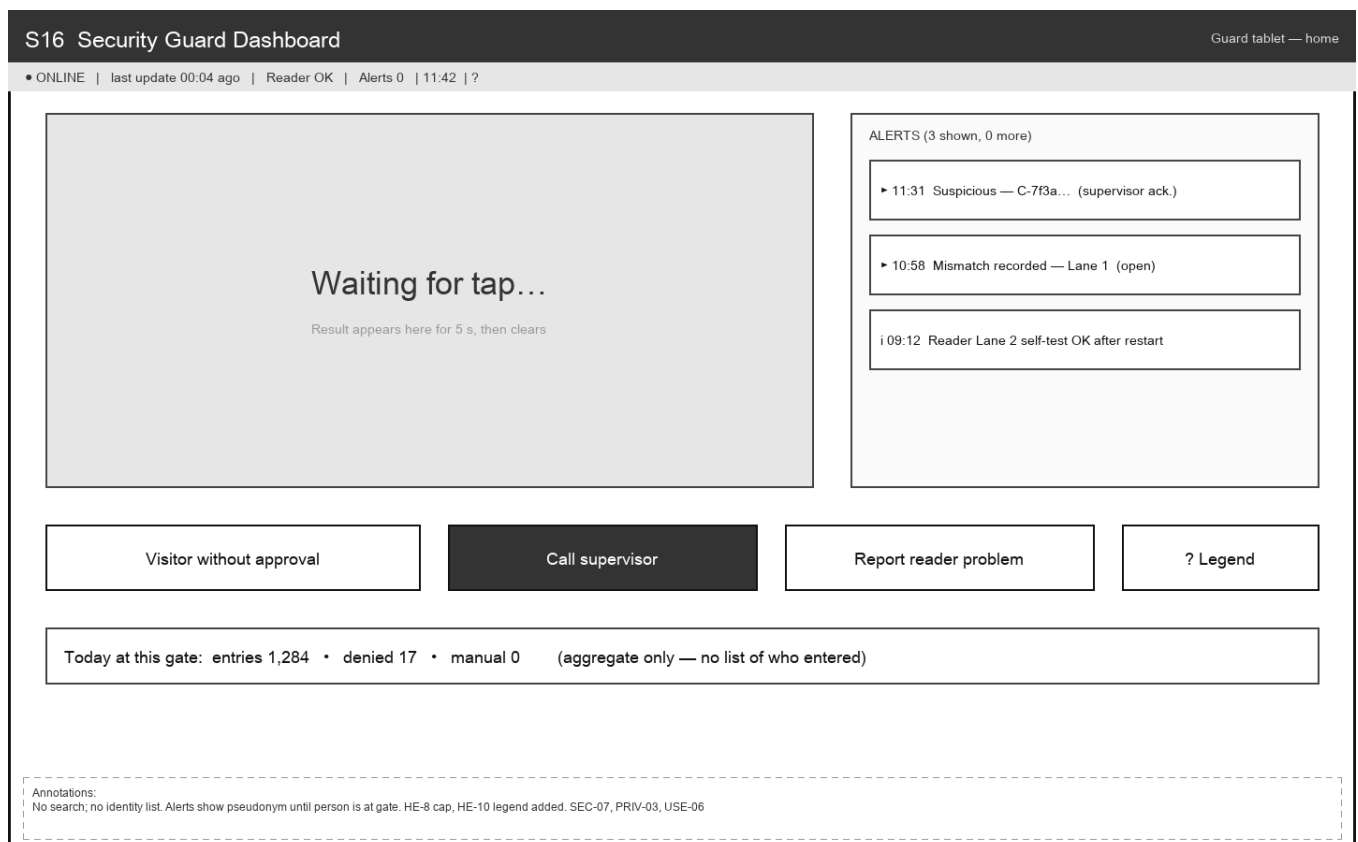
Annotations:

MFA session; decision logged. Visitor contact not shown (not needed to decide). SEC-12, PRIV-01

S14 — Visitor Approval



S15 — Temporary Visitor Credential



S16 — Guard Dashboard

S17 Admin Credential Management

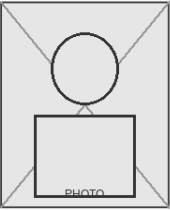
Issuer console

Role: ISSUER · MFA ✓ (TOTP) · session 14:59

Roster search

Meher S... / MT2026...

Search



Meher S...

Roll: MT2026...

Programme: M.Tech CSE

Role: STUDENT

Fields stored: only these.

No phone / address / hostel.

Issue new card

Send phone enrolment

Key material: present in HSM (never displayed)

CREDENTIALS

Card4821

REVOKED

lost — reported 28 Sep 2026

Card9130

ACTIVE

valid to 31 Jul 2028

Renew Suspend Revoke

Phone (Android)

ACTIVE

enrolled 28 Sep 2026

Suspend Revoke

CONFIRM: Revoke Card9130?

Reason: • Lost / Stolen / Disciplinary / Left

Effect: denied at all 6 gates within 30 s. Cannot be undone.

Cancel

Revoke permanently

Annotations:

Roster match required to issue; two-step confirm with exact effect; reason dropdown; all logged. Issuer = Auditor. FR-07, SEC-07, SEC-11, SEC-12, USE-10

S17 — Admin Credential Management

S18 Lost ID / Revoke Credential

Student app

My credentials

Card9130

ACTIVE

Tap to manage

Phone (this device)

ACTIVE

Report card lost or stolen?

• This will PERMANENTLY disable card9130.

• Your phone credential keeps working.

• A replacement request will be created.

• This cannot be undone.

Type DISABLE to confirm:

DISABLE

Report and disable

Cancel

Annotations:

Self-service, immediate, irreversible (HE-13: typed confirm).

After: card REVOKED, "Replacement requested — collect at ID cell with photo ID". FR-09, SEC-17, USE-10

S18 — Lost ID Revoke

S19 Entry Logs

Auditor console

Role: AUDITOR • MFA ✓ • Chain verified ✓ (last checkpoint 12:00)

Filters:

Gate • Main

From • 01 Oct 00:00

To • 01 Oct 12:00

Decision • All

Method • All

⌵ Flagged only

Time	Gate	Method	Decision	Credential (pseudonym)	Flags	Guard ID	
11:31:02	Main	card	DENY-SUSPICIOUS	C-7f3a9c...	▶ velocity	—	Re-identify
10:58:41	Main	card	DENY-SUSPICIOUS	C-21be0d...	▶ mismatch	G-14	Re-identify
09:40:17	Main	manual	ALLOW (manual)	— (type: card)	▶ manual	G-14 / SUP-07	Re-identify
08:52:09	East	QR	DENY-REVOKED	C-9a0f31...	▶ revoked	—	Re-identify
08:51:55	Main	NFC	ALLOW	C-0cd7e2...		—	Re-identify

No edit or delete controls exist on this screen. Export is pseudonymous.

RE-IDENTIFY C-7f3a9c...

Case reference (required)

SEC-2026-0142

Justification (required)

! This action is logged and visible to the second administrator.

Cancel

Confirm

Annotations:
 Pseudonymous by default; re-identification justified + logged + visible to second admin; chain integrity indicator. SEC-10, SEC-15, PRIV-04, PRIV-06, FR-15

S19 — Entry Logs

S20 Privacy / Access-Control

Student app (left) • Policy admin (right)

STUDENT — Privacy & my data

What is collected
 name, roll no., programme, photo, role, credential status, entry events (pseudonymous)

Why
 access decisions; security incidents; system integrity. NOT attendance or tracking.

Who can see it
 Guard: photo + first name + role, only when you tap. Auditor: pseudonymous; re-identify only with a case ref.

How long
 entry events 90 days; flagged events until case closed + 1 year

My entry records (last 90 days)
 01 Oct 08:51 Main NFC ALLOW • 30 Sep 21:14 Main card ALLOW • ...

My credentials
 Card ---9130 ACTIVE • Phone ACTIVE • [Request correction] [Full notice]

POLICY ADMIN — Gate access policy (v12, edited by PA-02, 29 Sep)

Gate	Students	Staff	Visitors (pass)	Contractors
Main	06:00–23:00	24 h	pass window	08:00–18:00
East	06:00–22:00	24 h	pass window	DENY
Vehicle	DENY	24 h	pass window	08:00–18:00
Any other role	DENY	DENY	DENY	DENY

Last row is the secure default and cannot be edited.

CONFIRM policy change

Gate: East • Role: Students

Before: 06:00–22:00 After: 06:00–23:00

Applies to 2 readers. MFA step-up required.

Change is versioned and logged (v13).

Cancel

Apply (MFA)

Annotations:
 Transparency + own-data access (DPDP). Deny-by-default made visible. Policy edits: MFA, before/after, versioned. PRIV-07, PRIV-08, FR-18, SEC-11, DP-3

S20 — Privacy Access Control